PalAST

Faculty of Information Technology - IUG

A Mobile Application for Automatic Antimicrobial Susceptibility Testing

Team Members

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Supervisor

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Presentation Outline

PALAST

1

Introduction

Briefly elaborate on what you want to discuss.

Problem

- Antibiotic resistance is a growing problem worldwide.
- One way to combat it is through antibiotic susceptibility testing (AST).

AST

Antibiotic Susceptibility Testing (AST) is a test that aims to identify the most effective antibiotics for treating bacterial infections.

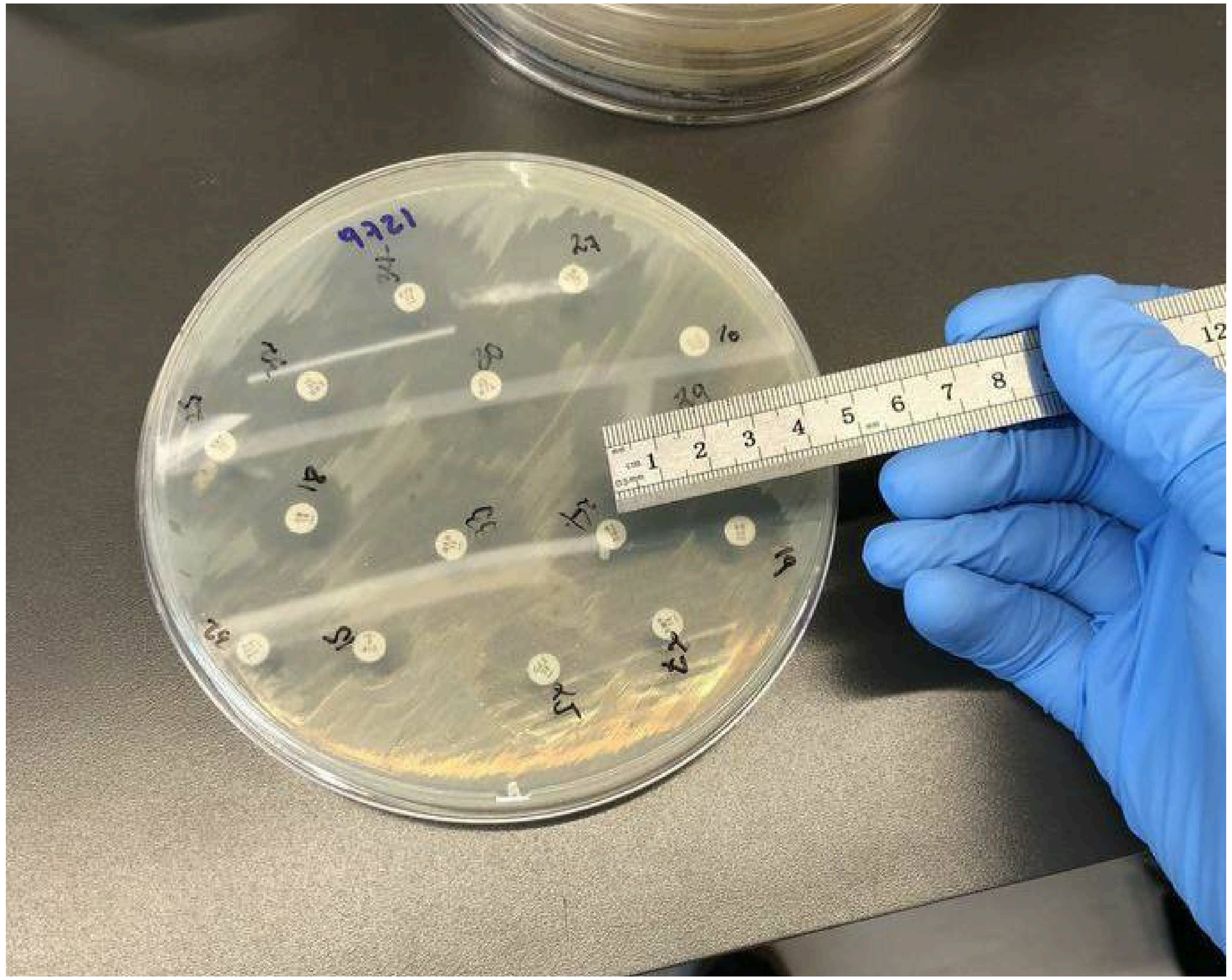
AST Test Types

There are 2 ways to carry out an AST disc diffusion:

Manual methods	Automated methods

Manual Methods

- Less expensive
- Performed in resource-limited settings
- More time-consuming and prone to human error.



Automated Methods

- Faster
- More accurate
- Require specialized equipment and can be more expensive to set up and maintain.

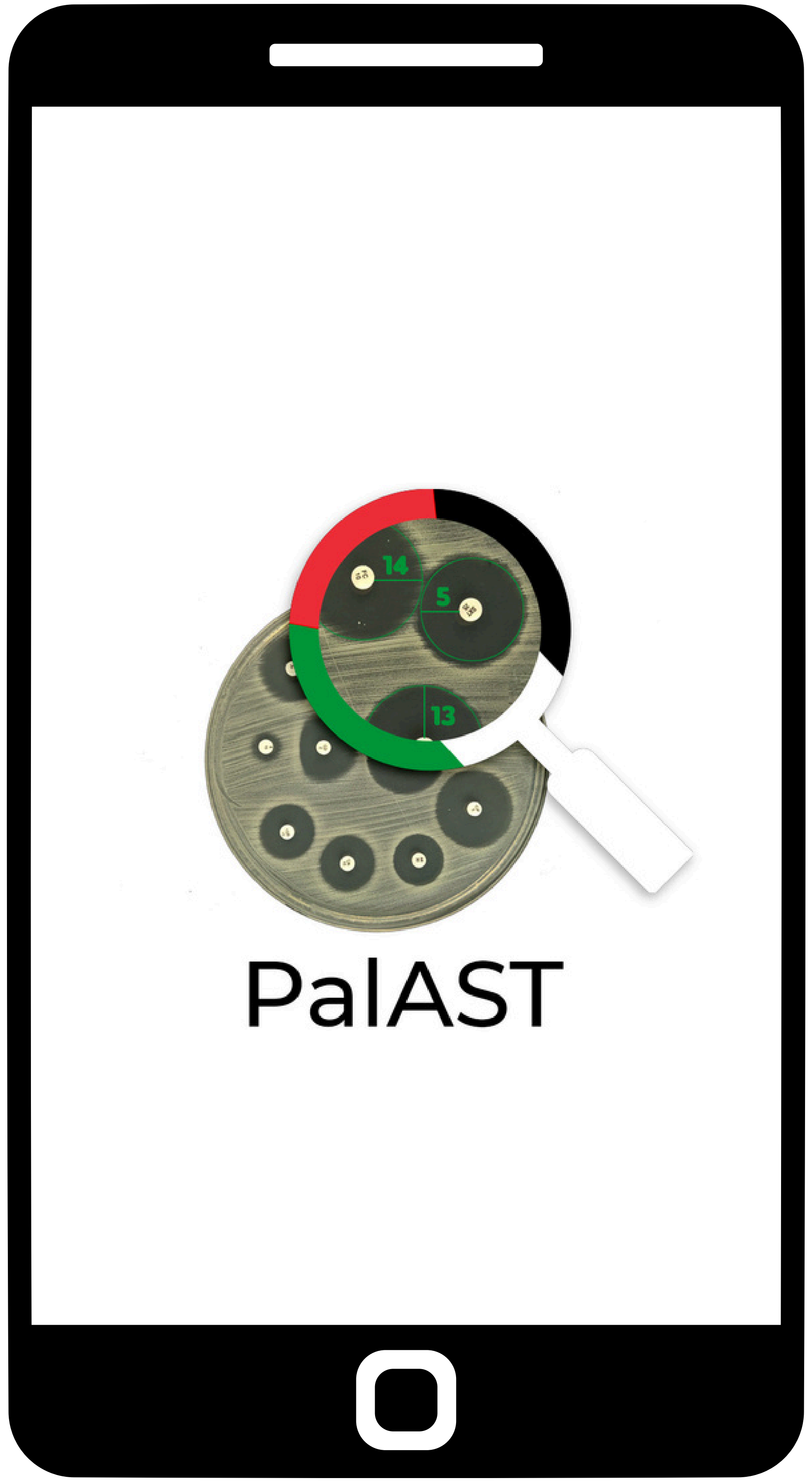
AST Test Types →

Increasing antimicrobial resistance caused a need for an alternative solution to improve the

- Speed
- Accuracy
- Standardization of AST, and
- Reduce costs

Solution!

- A mobile application that perform AST automatically
- Faster, easier, less expensive, and more accurate!



2 Related Works

Briefly elaborate on what you want to discuss.

AntibiogramJ

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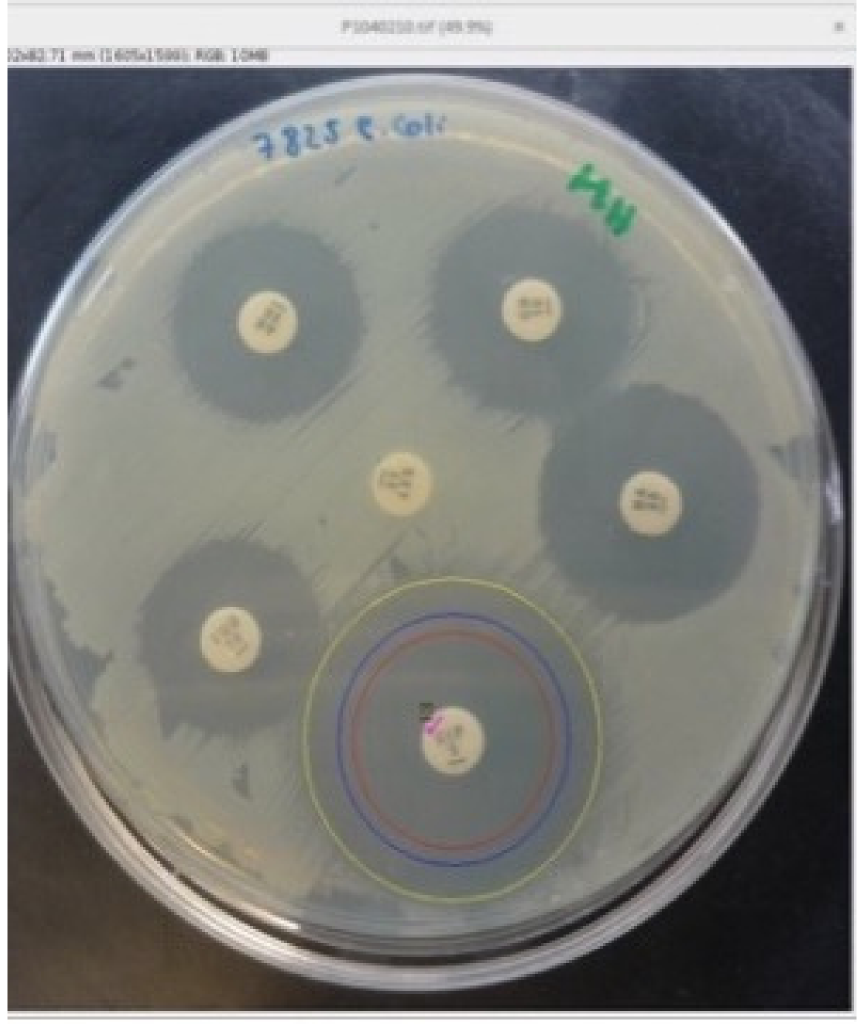
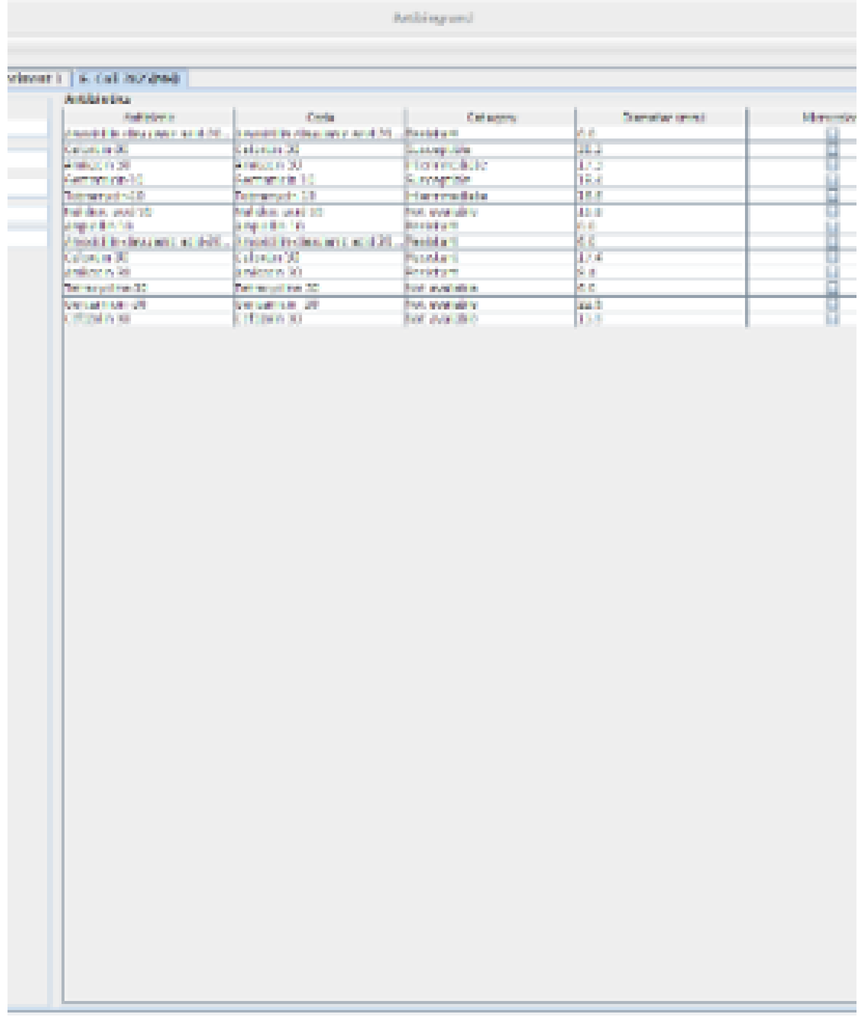
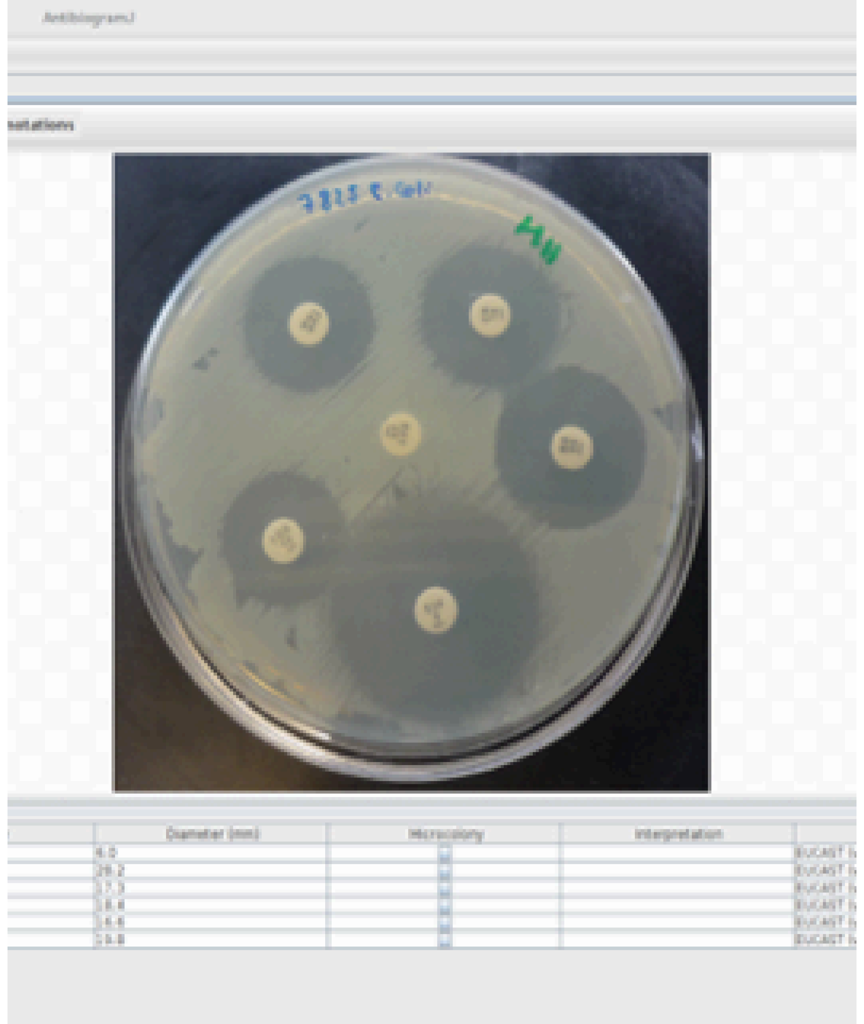
AntibiogramJ is a desktop Java-based application that takes antibiogram images as input, then identifies and measures inhibition zones as output.







Baltekin, et al.

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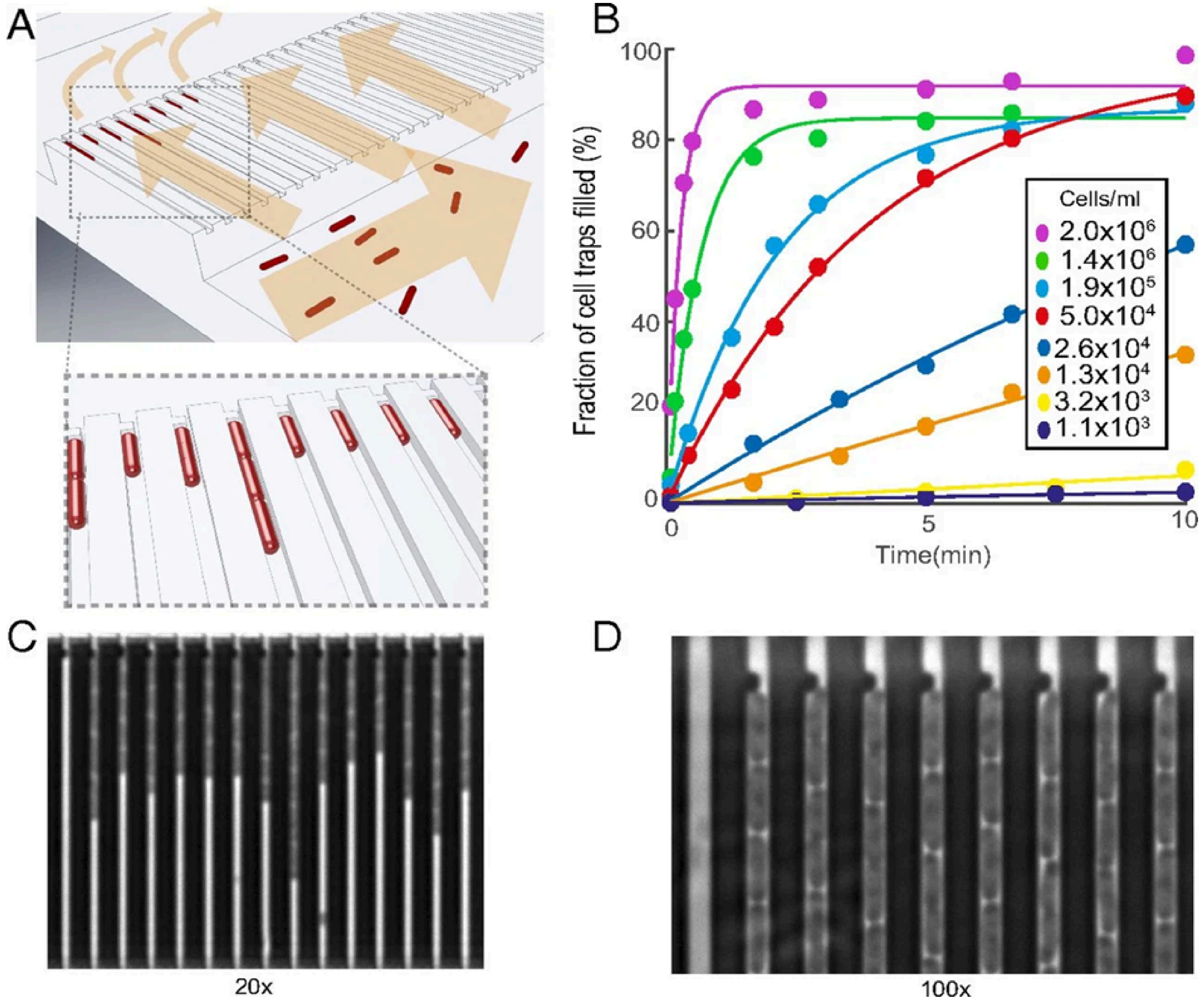
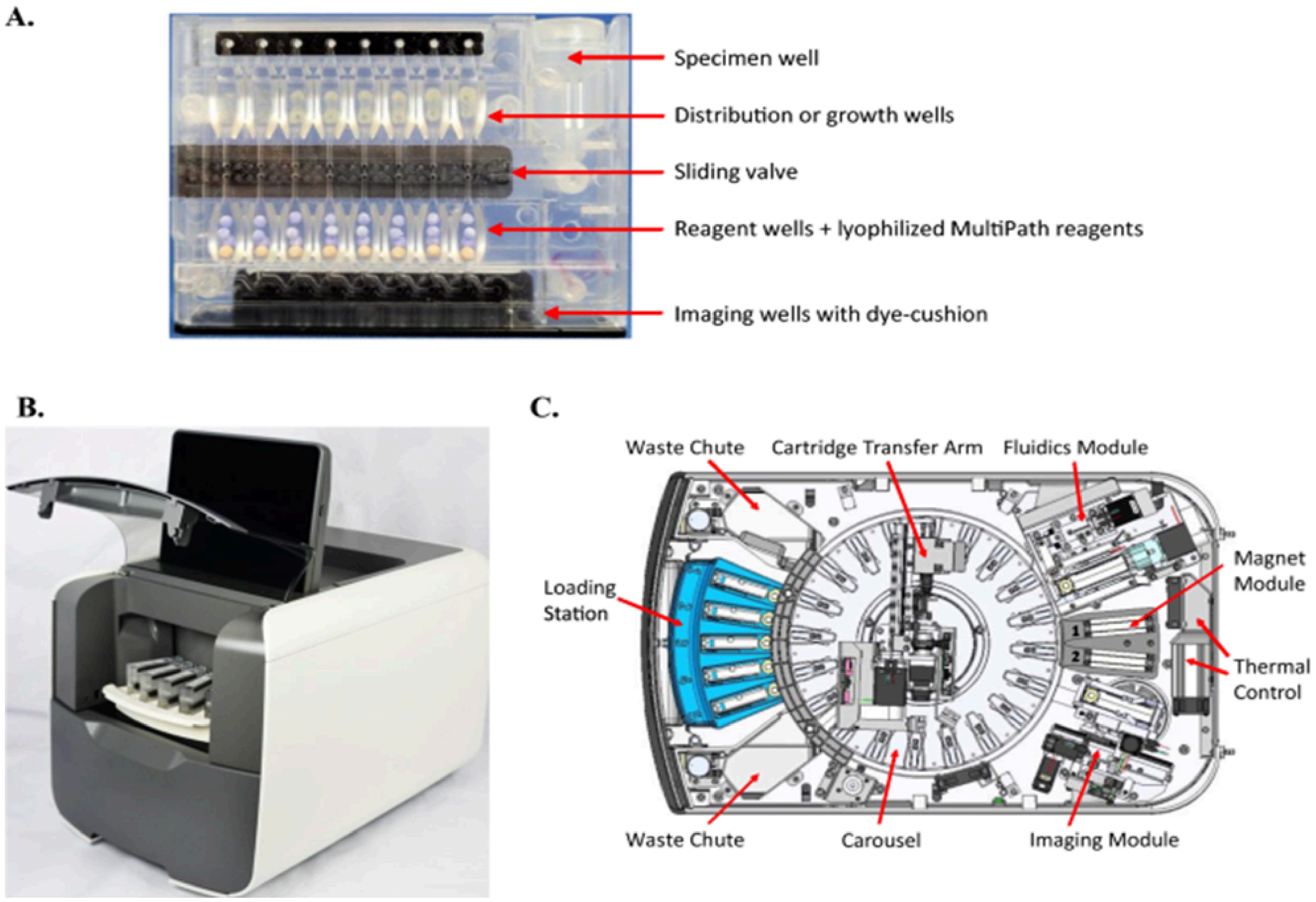
Baltekin, et al. developed a point-of-care susceptibility test for urinary tract infection.



Burg, et al.

→

Burg, et al. proposed an automated platform that identifies urinary tract infection pathogens.



Panpradist, et al.

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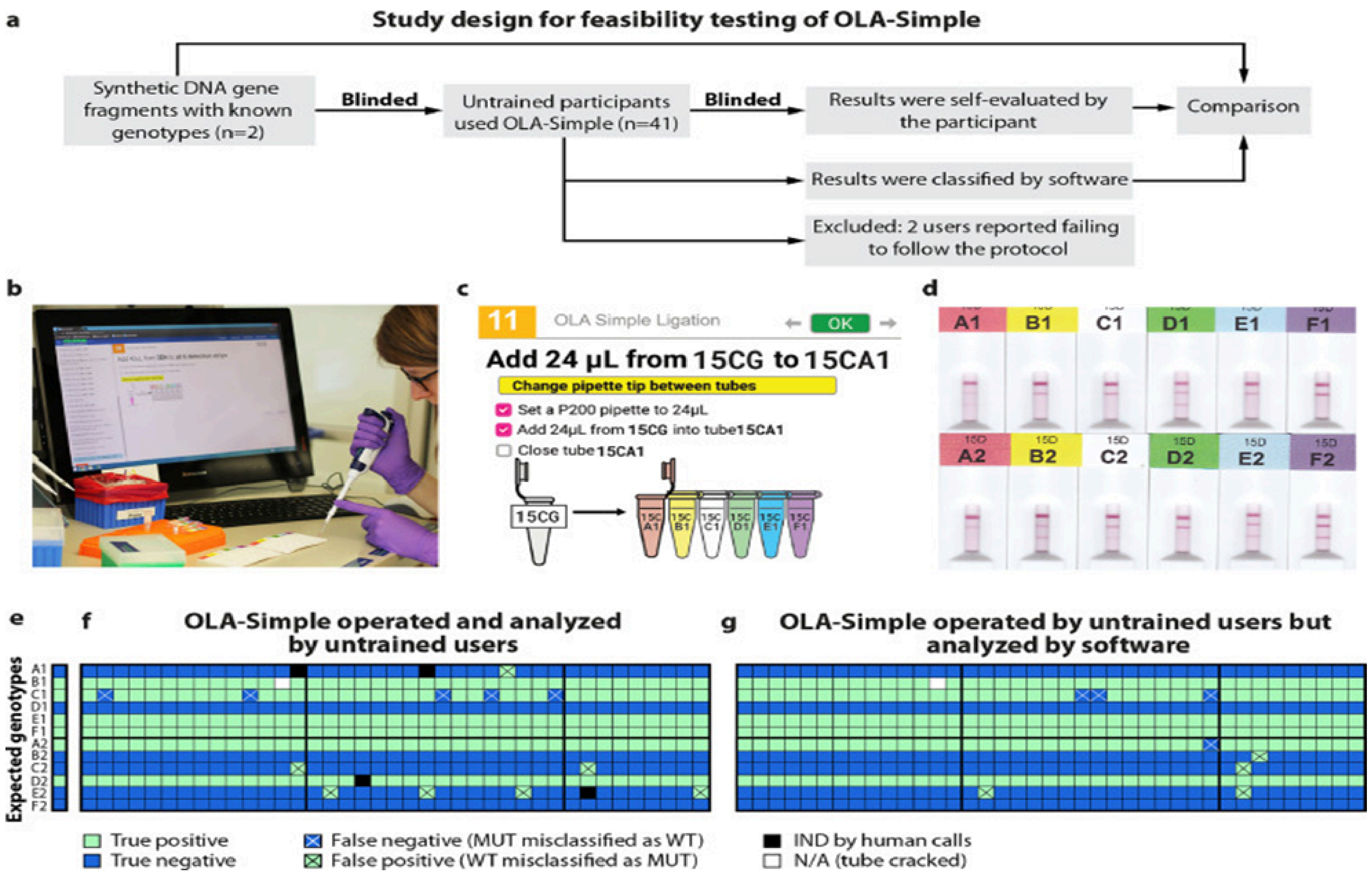
Panpradist, et al. proposed OLA (oligonucleotide ligation assay), which is a kit developed for detection of HIVDR.



Pascucci, et al.

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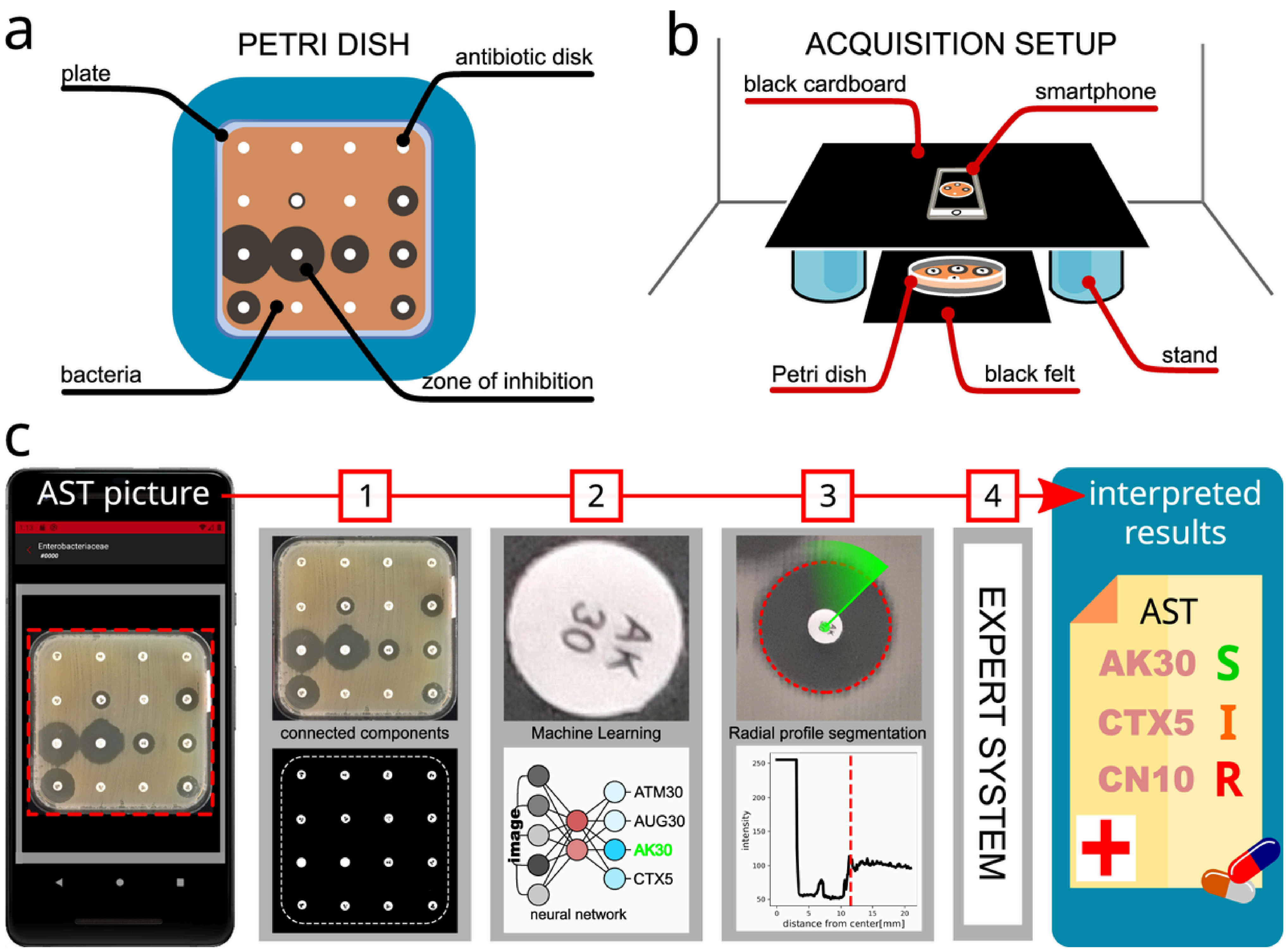
Pascussi et al presented an AI-based mobile app for antimicrobial analysis.

It's not publicly available.



Comparison between works that tackle automated AST

Work	Measurement of inhibition zones	Identification of antibiotic labels	Interpretation of measurements	Mobile App
AntibiogramJ	✓	✓	✓	
Baltekin, et al.	✓ (For urinary tract infection only, requires custom-designed chip)	✓	✓	
Burg, et al.	✓ (For urinary tract infection only, requires special devices)	✓	✓	

Comparison between works that tackle automated AST

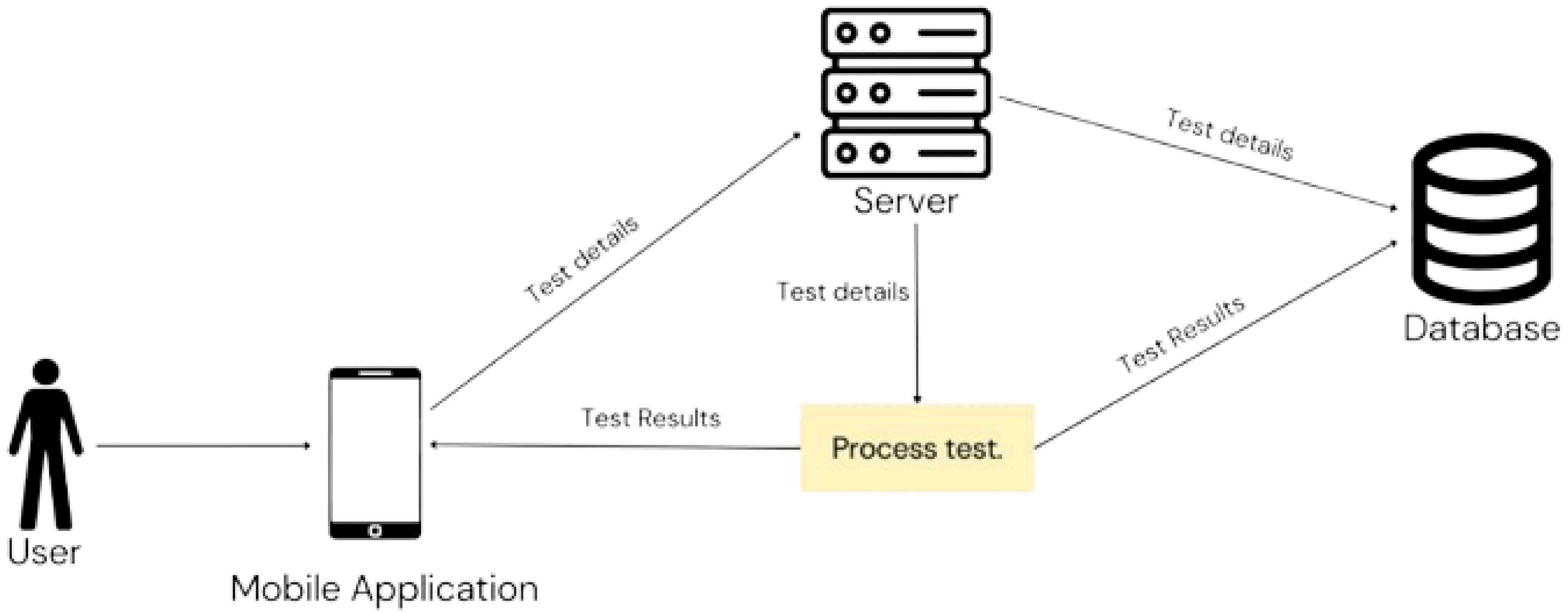
Work	Measurement of inhibition zones	Identification of antibiotic labels	Interpretation of measurements	Mobile App
Panpradist, et al	✓ (a kit for detection of HIV Drug Resistance)	✓	✓	
Pascucci, et al	✓	✓	✓	✓ (Not available for public use, only for Android device)
PalAST	✓	✓	✓	✓ (cross-platform)

3

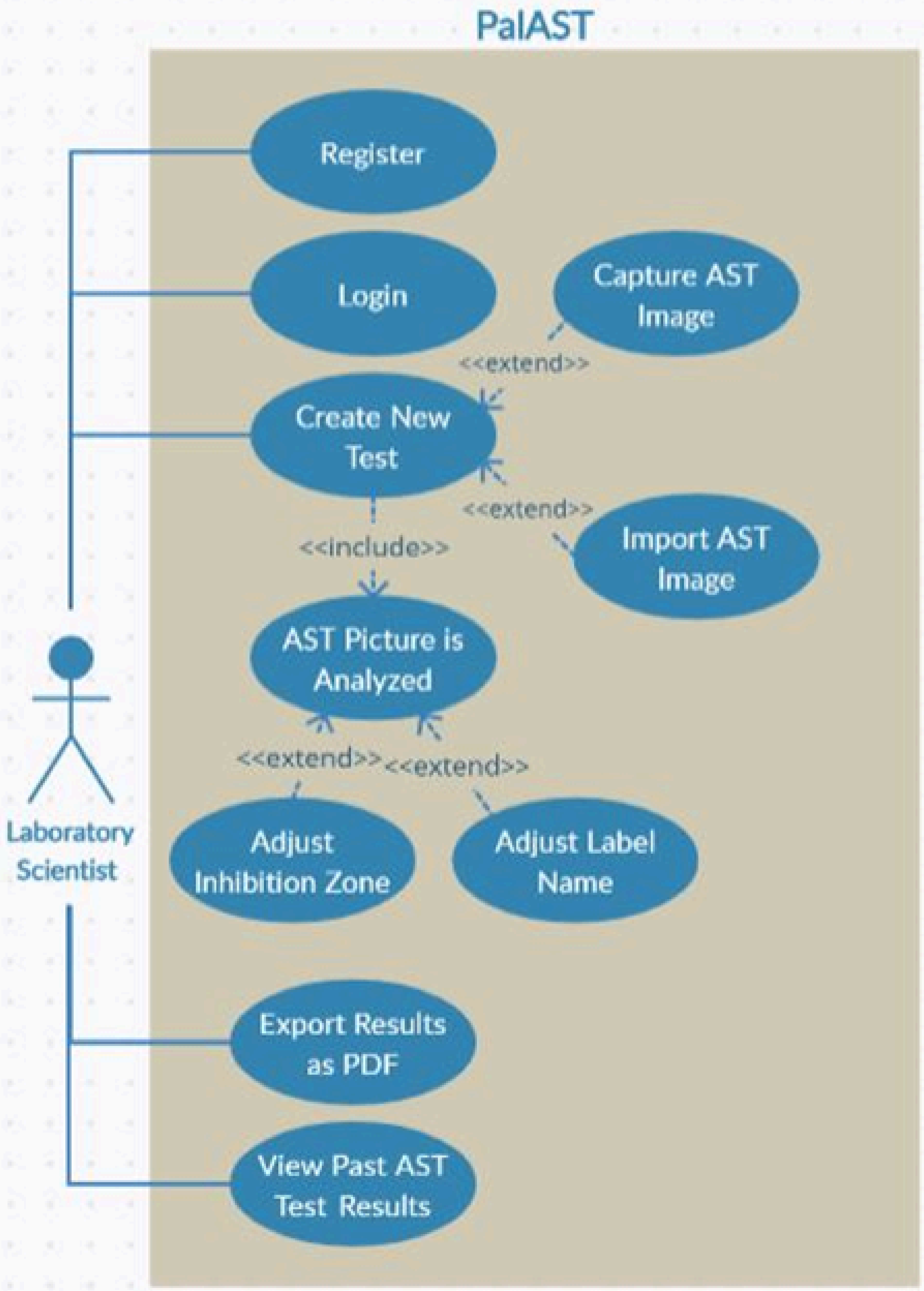
Design

Briefly elaborate on what you want to discuss.

High Level System Design



PALAST



Use Case Scenarios

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This diagram represents the use case scenarios for PalAST

4

Implementation

Briefly elaborate on what you want to discuss.

Implementation

PalAST has two components:

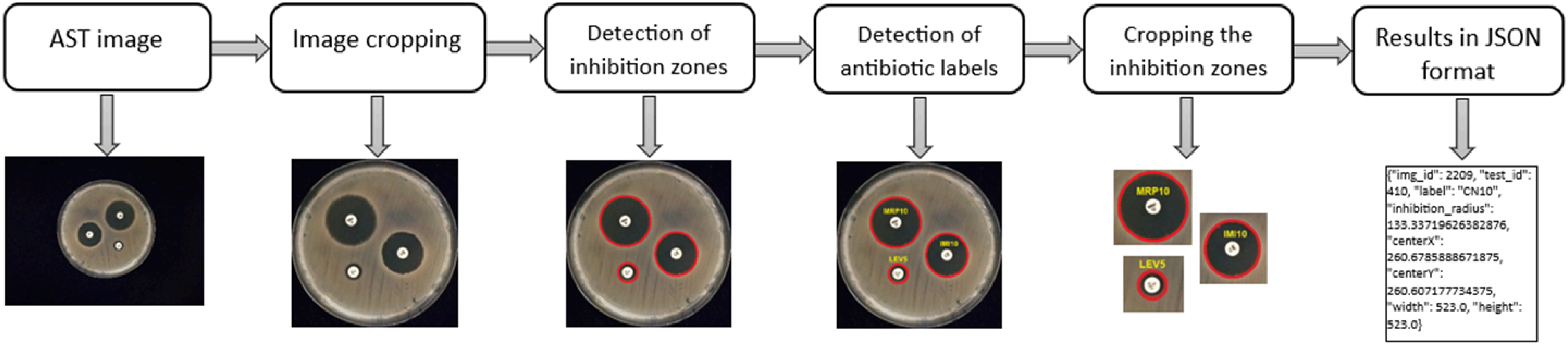
The Server Side

- Handles all the AI and image processing functionalities of PalAST.

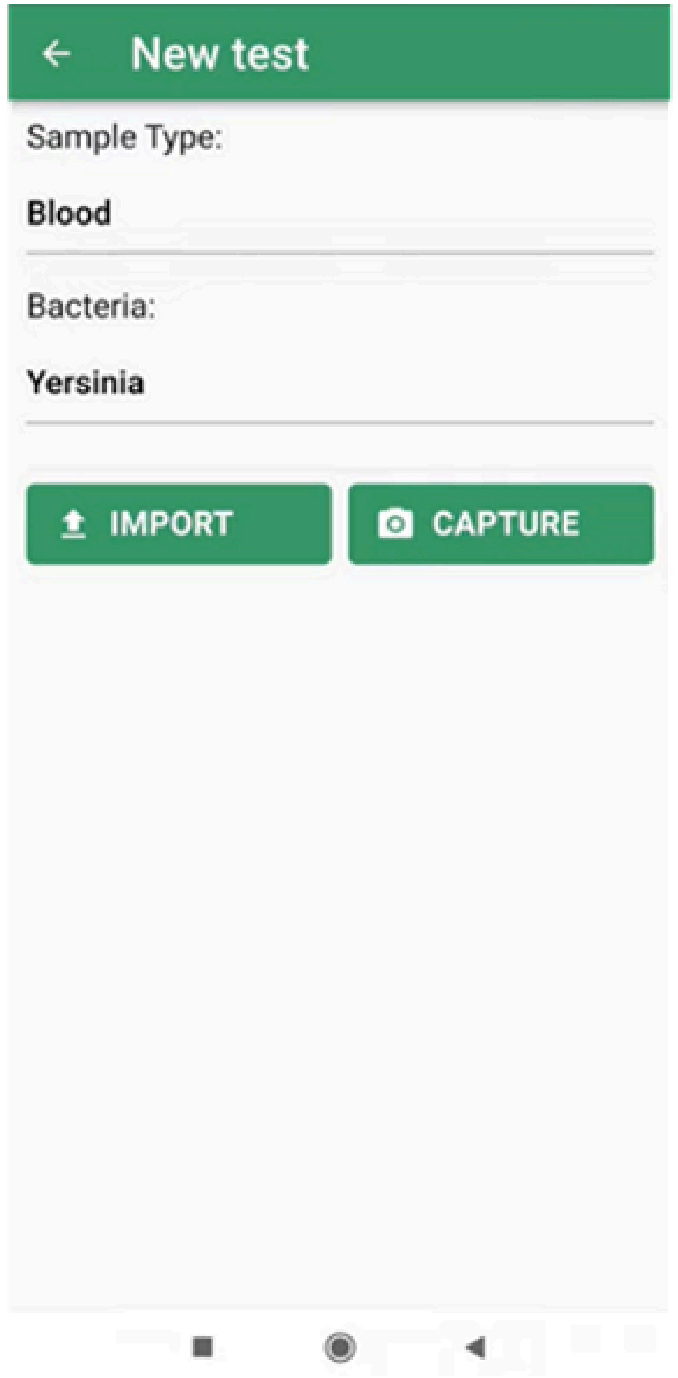
The Client Side

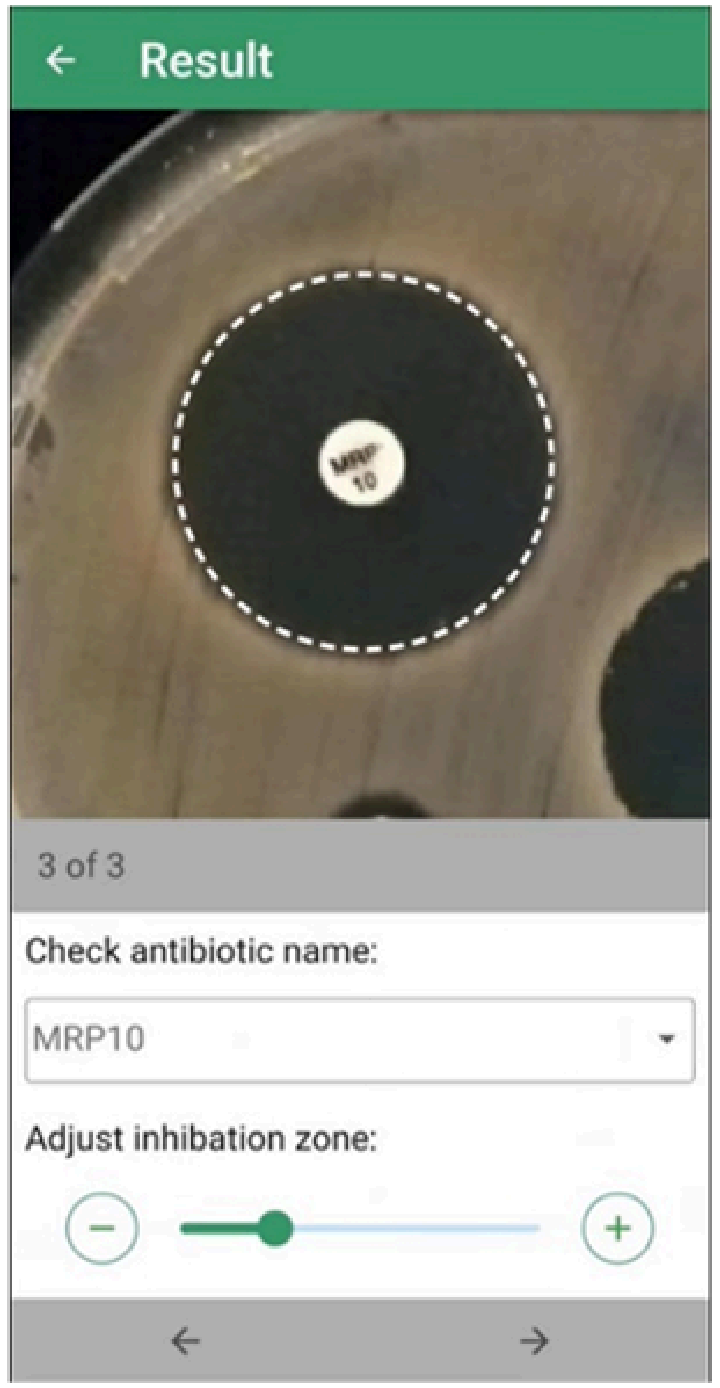
- Responsible for displaying results and communicating with the server through a RESTful web service.

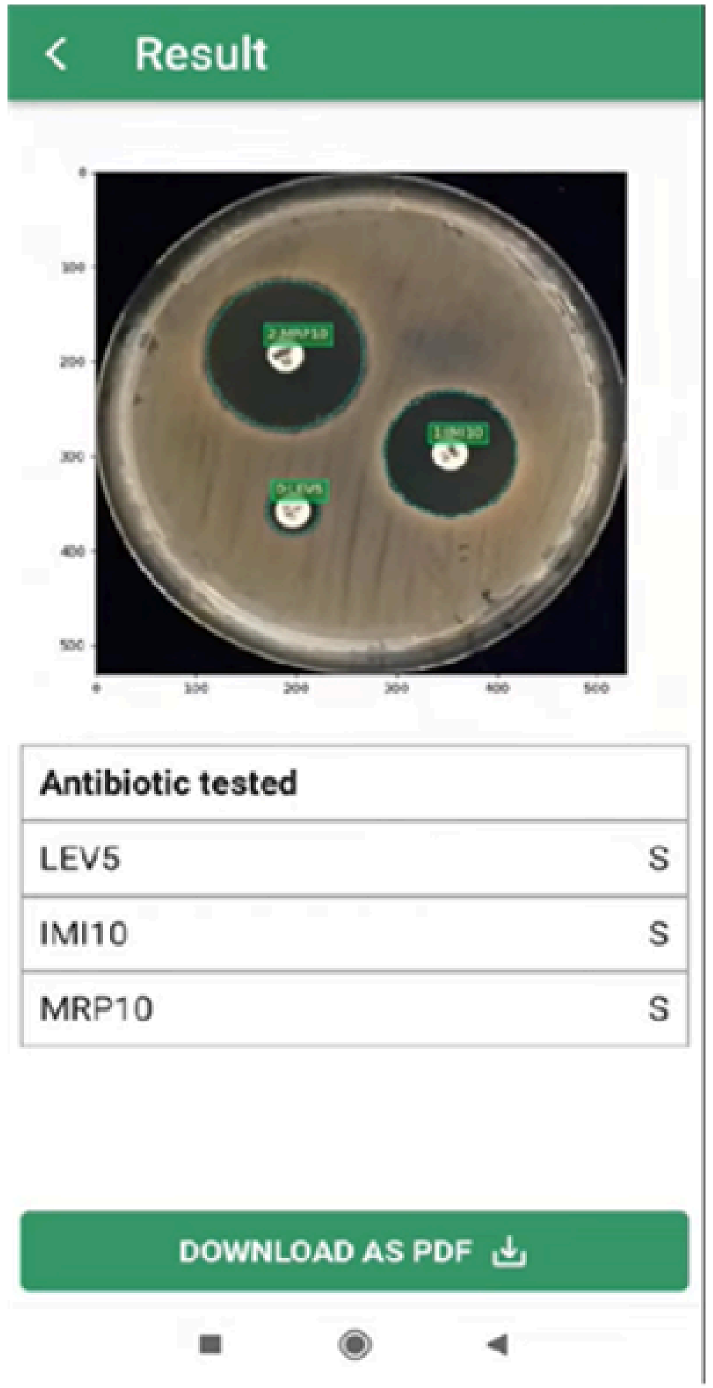
PalAST in Action

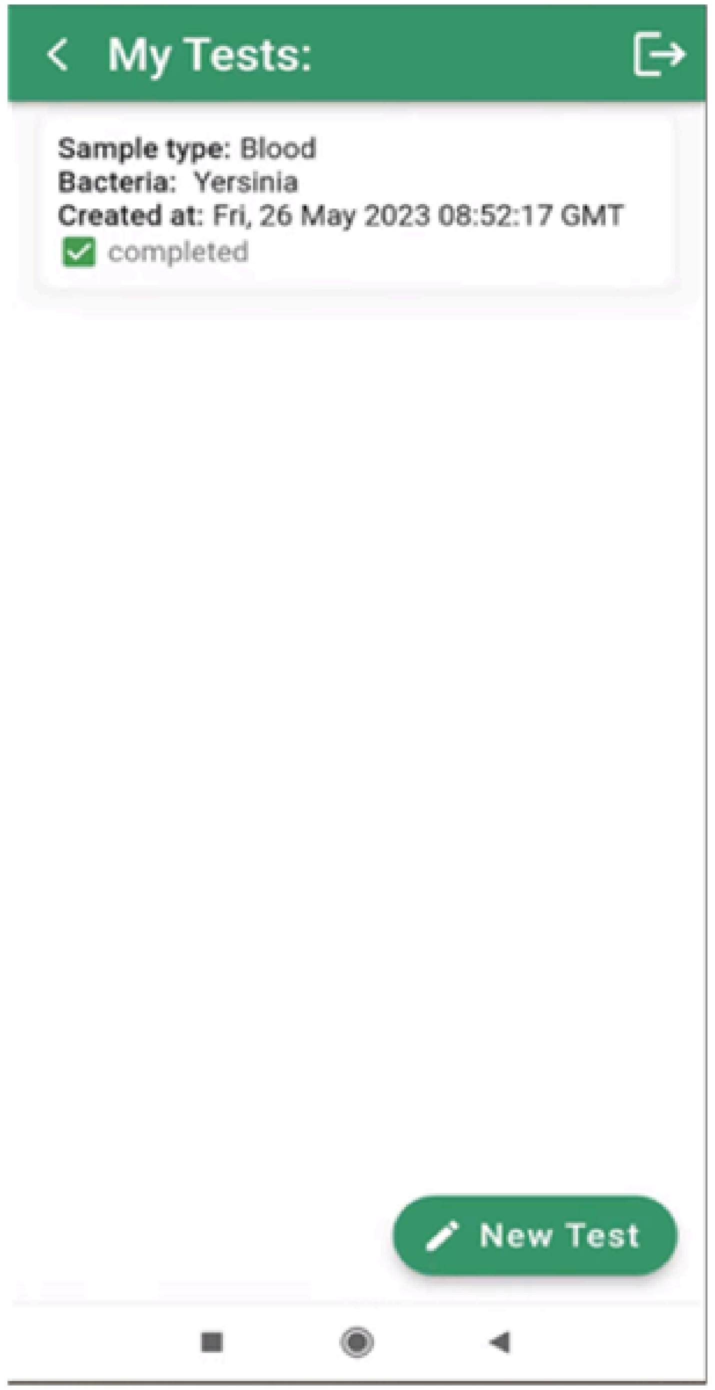


Screenshots of sample screens from PalAST







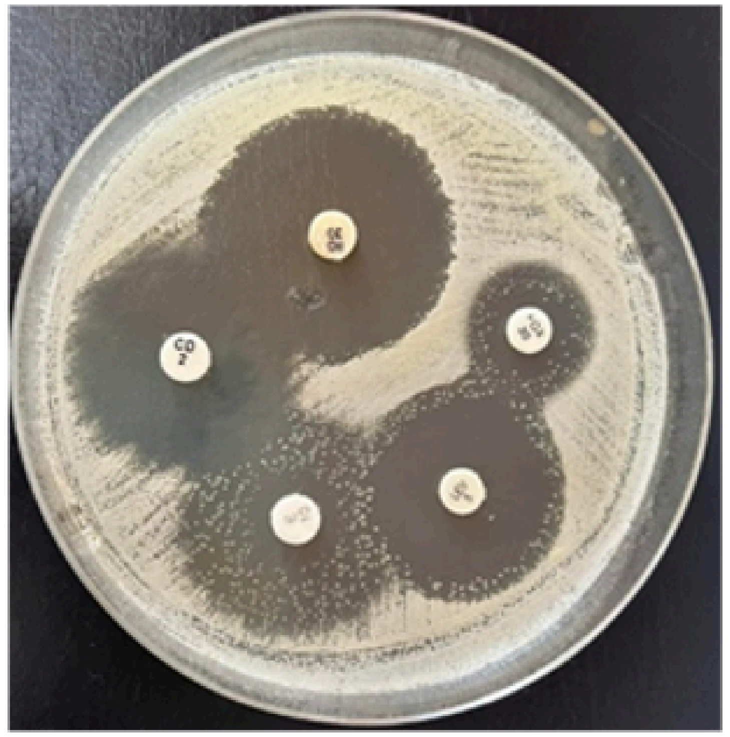
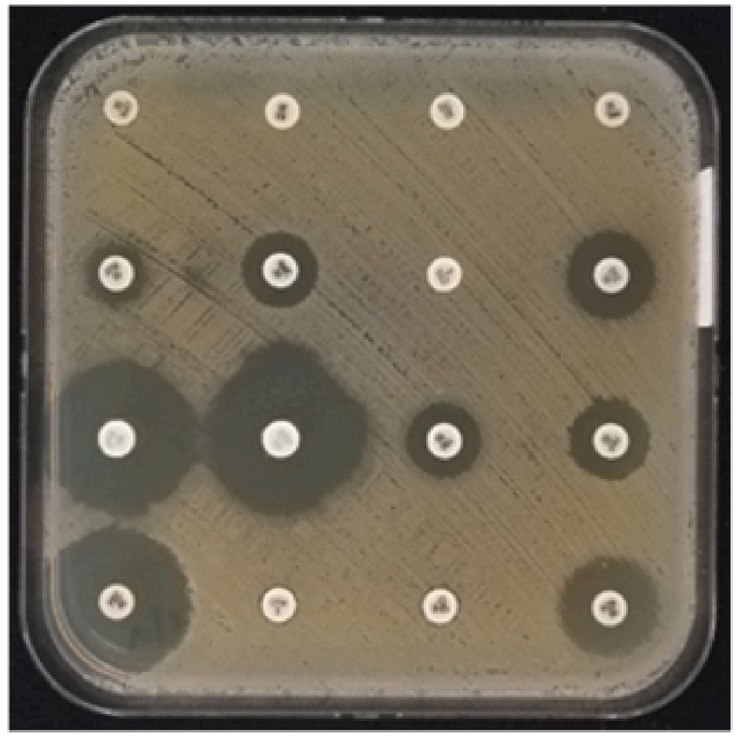
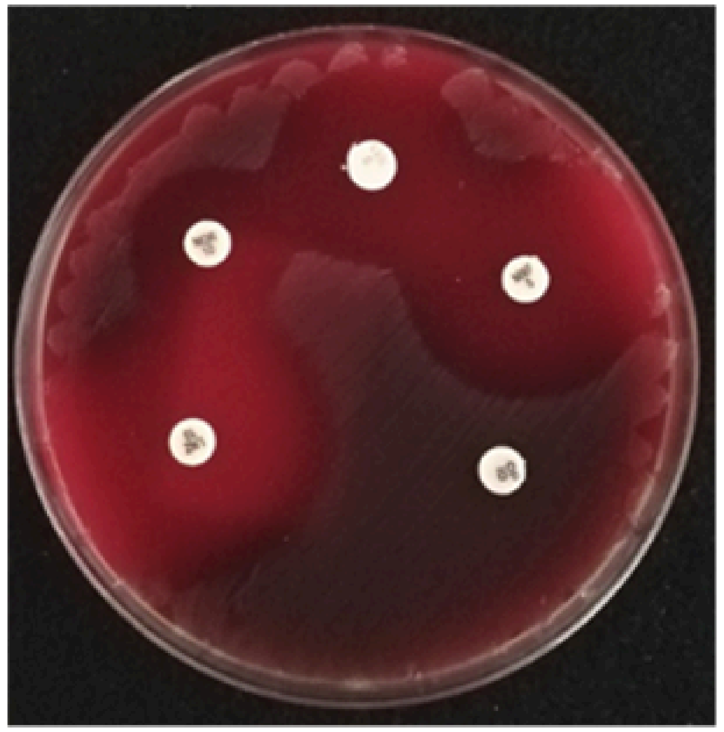
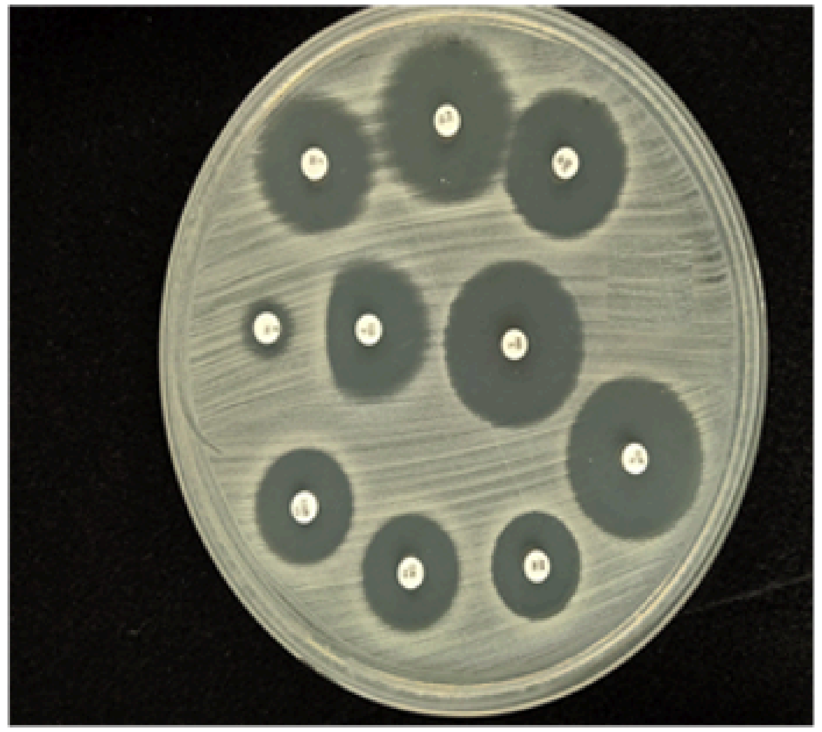


5 Evaluation

Briefly elaborate on what you want to discuss.

Data Collection

- We used sample photos of agar plates to evaluate PalAST
- The samples include various types of bacteria and antibiotics

Evaluation of PalAST’s Performance

Two evaluation methods were used:

Common metrics for evaluating object detection.	Expert evaluation

Common Metrics for Evaluating Object Detection

	Zone Detection	Antibiotic Label Detection
Precision	0.858	0.733
Recall	0.795	0.810
IoU	0.706	Not Applicable

EXPERTS' EVALUATION

Two experts evaluated PalAST using a 5-point Likert scale (shown in next slide). Both experts :

- Agreed that PalAST saves time and effort.
- Liked the fast and automatic AST analysis.
- Suggested adding new antibiotics and bacteria species and updating the interpretive criteria of AST.
- Highlighted the need to run the application offline.

EXPERTS' EVALUATION

Question	Expert 1 responses	Expert 2 responses
PalAST saves time and effort by automating the AST	Very good	Very good
PalAST enables me to easily start a new AST test and capture AST photo	Very good	Good
Inhibition zones are accurately detected and measured by PalAST	Good	Good
Antibiotic names are accurately detected and recognized by PalAST	Good	Good
PalAST responds quickly to AST requests and presents results without incurring significant delay	Very good	Very good
PalAST enables me to easily verify the analysis results and correct any wrong detections or measurements.	Good	Good
PalAST correctly rates bacteria as susceptible, intermediate, or resistant to each antibiotic	Good	Good
PalAST enables me to access and review past test results	Very good	Very good
PalAST is easy to use and navigate	Very good	Very good

6 Future Work

Briefly elaborate on what you want to discuss.

THE FUTURE OF PALAST

In the future we aim to upgrade PalAST to:

- Work offline
- Conduct a large-scale evaluation of PalAST
- Be organization oriented

https://www.youtube.com/watch?v=kaymrFW9rgo&feature=youtu.be

PalAST'S YouTube Tutorial

ACADEMIC YEAR OF 2022-2023

Thank you!